Maynooth University
National University of Ireland Maynooth

SEMESTER 1
2023-2024

CS264FZ
Software Design

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Time allowed: 2 hours

Answer all ***four*** questions

All questions carry equal marks

Instructions

	Yes	No
Log Books Allowed	☐	☑
Formula Tables Allowed	☐	☑
Other Allowed (*enter details)*	☐	☑

General (*enter details)*

QUESTION 1

(a) Please give an example to explain how the state pattern is used in the real world. (5 marks)

(b) Can extrinsic data of flyweights be shareable? How do clients handle the data? (5 marks)

(c) When should we use the interpreter pattern? Please give an example. (5 marks)

(d) Can we use different types of observers in the same application? Please give your reason. (5 marks)

(e) How are the abstraction and implementation implemented in the bridge pattern? You can use classes ElectronicGoods and Sate in the textbook as an example. (5 marks)

QUESTION 2

(a) Suppose there is an interface Employee and its two concrete subclasses, Worker and Manager. Which pattern can be used to add a new operation, such as getSalary(), to classes Worker and Manager without modifying them? Please give your solution. (5 marks)

(b) Please describe the differences and relationships between the mediator and observer patterns. (5 marks)

(c) Please explain the differences between the eager and lazy initializations in the singleton pattern. (5 marks)

(d) Can the iterator pattern be used with the composite pattern? Please give your reason. (5 marks)

(e) Please describe a solution to handle the situation in the chain-of-responsibility pattern where the request cannot be handled by any of the handlers in the chain. (5 marks)

QUESTION 3

(a) It is necessary to create character classes and strategy interfaces when the strategy pattern is used to simulate attacks of different characters in a game. The Test class and the program's output are as follows：

```
// this is some code
public class StrategyPatternExample {
    public static void main(String[] args) {
        Character player = new Character();
        Character enemy = new Character();
        AttackStrategy magicalAttack = new
MagicalAttackStrategy ();
        AttackStrategy physicalAttack = new
PhysicalAttackStrategy ();
        player.setAttackStrategy(magicalAttack);
enemy.setAttackStrategy(physicalAttack);
        player.attack(enemy);
        enemy.attack(player);
    }
}
```

Outputs:
magical attack
physical attack
Please finish the following tasks.

(i) Create an attack strategy interface AttackStrategy with an attack() method. (5 marks)

(ii) Create two attack strategy implementation classes, namely MagicalAttackStrategy and PhysicalAttackStrategy. (10 marks)

(iii) Create a character class Character to implement different attack strategies. (10 marks)

QUESTION 4

(a) Suppose you are designing an e-commerce platform that needs to integrate multiple third-party payment gateways, including Alipay, WeChatPay, and UnionPay. Please use the facade pattern to design a unified payment interface, allowing customers to choose and complete payment operations through a simple interface. The detailed requirements are as follows:

(i) Create three classes, namely Alipay, WechatPay, and UnionPay, to implement the three payment gateways, respectively. Each class contains a payment() method. (8 marks)

(ii) Write a facade class PaymentFacade to encapsulate the payment operations of the three payment classes. (12 marks)

(iii) Write a Test class to test the functions of the above classes. (5 marks)